

A Intervention-targeted lineage perturbations

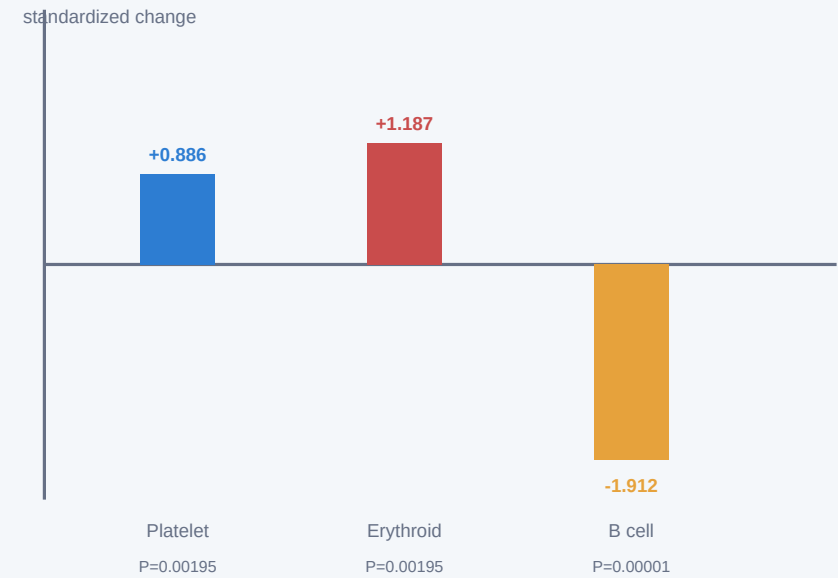
Eltrombopag ITP → Platelet restoration

Erythropoietin Healthy volunteers → Erythroid restoration

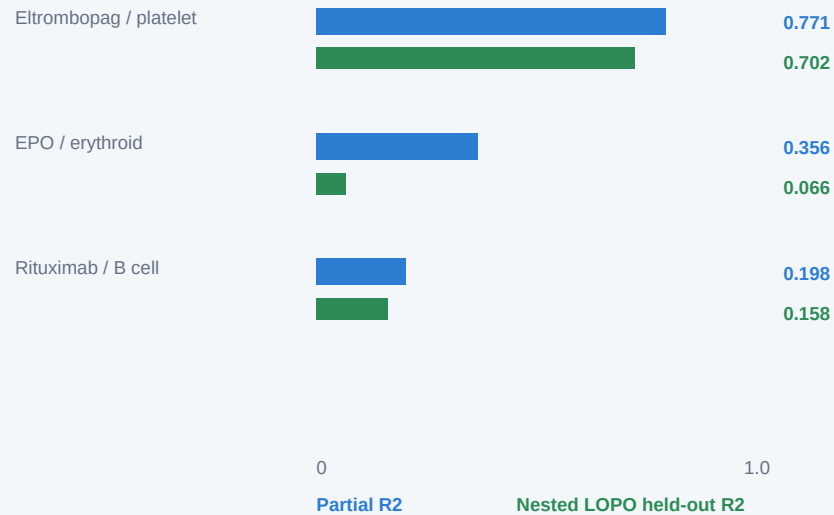
Rituximab Type 1 diabetes → B-cell depletion

Externally specified scores; no outcome-driven marker selection

B Target-lineage RNA-score changes



C Variance attribution and held-out reconstruction



D Composition-only benchmark (100 replicates)



Scores are lineage-derived RNA-abundance proxies, not measured cell counts.